Therapeutic uses of local hot applications

- Heat decreases pain
- Heat decreases muscle tone
- Heat promotes healing
- Heat promotes suppuration
- Heat relieves deep suppuration
- Heat provides warmth
- Heat stimulates peristalsis

Therapeutic uses of local cold applications

- Cold relieves pain
- Prevents gangrene
- Prevents edema and reduce inflammation
- Controls hemorrhages
- Checks the growth of bacteria
- Reduce the body temperature
- Cold anaesthetize an area

Principles of hot and cold applicatons

1. Water is good conductor of heat.
2. Air is poor conductor of heat.
3. Heat always flows from hotter area to the less hot area.
4. Prolong exposure to moisture increases the skin susceptibility to maceration and skin breakdown, reducing the protection of the intact skin.
5. Moisture left on the skin cause rapid cooling due to evaporation of the moisture.
6. Presence of steam increases the temperature of the hot application
7. Oil acts as the insulator and delays the transmission of the heat.
8. Woolen materials absorb moisture slowly, but hold the moisture longer and cold off less quickly than the cotton materials.
9. When immersed in water the body becomes buoyant therefore the exercises are performed under water with less effort.
10. The temperature tolerance varies with individuals and according to the site and area covered.
11. The end organs of the sensory nerves of the skin convey the sensation of cold, heat pain and pressure. The sensation is interpreted in the brain.
12. Friction produces heat.

Contraindications of hot applications

- Heat is not used in malignancies
- Heat is not used in patients with heart, kidney and lung diseases
- Should not used in acute inflamed areas.
- Should not be applied on patients with paralysis.
- Should not be applied on open wounds
- Should not be applied when there is an edema associated with venous or lymphatic diseases.
- Should not be applied on patients with metabolic disorders.

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